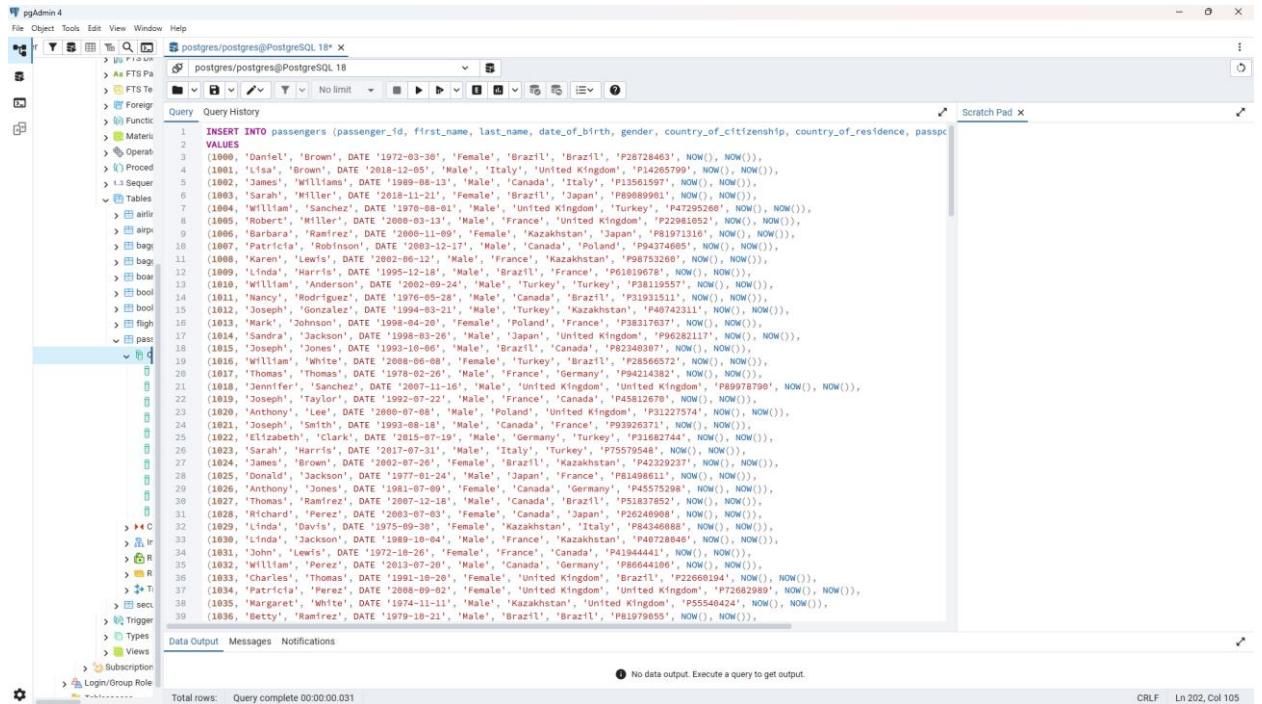


Laboratory work 3 Akhan Arman

1.

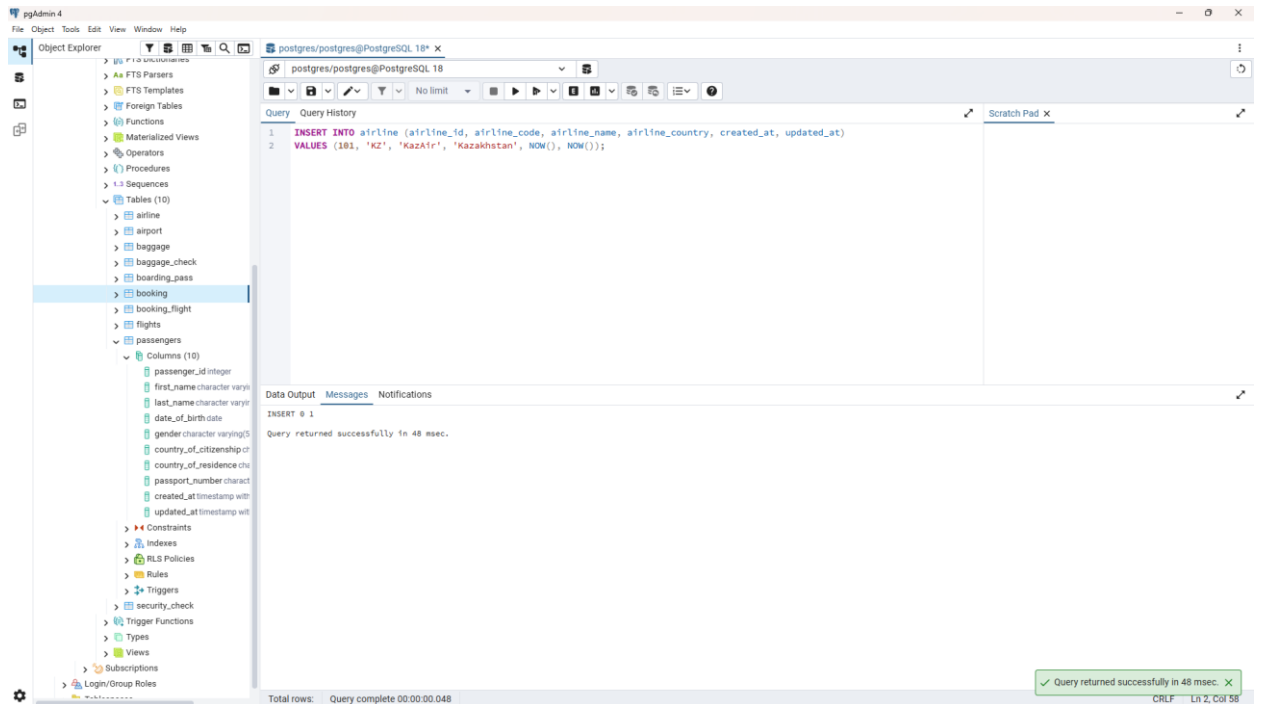


The screenshot shows the pgAdmin 4 interface. The 'Query History' window is open, displaying a large SQL query that inserts 48 rows into the 'passengers' table. The query is as follows:

```
1 INSERT INTO passengers (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, updated_at)
2 VALUES
3 (1000, 'Daniel', 'Brown', DATE '1972-03-30', 'Female', 'Brazil', 'Brazil', 'P28728463', NOW(), NOW()),
4 (1001, 'Lisa', 'Brown', DATE '2018-12-05', 'Male', 'Italy', 'United Kingdom', 'P14285799', NOW(), NOW()),
5 (1002, 'James', 'Williams', DATE '1989-08-13', 'Male', 'Canada', 'Italy', 'P13561597', NOW(), NOW()),
6 (1003, 'Sarah', 'Miller', DATE '2018-11-21', 'Female', 'Brazil', 'Japan', 'P89689981', NOW(), NOW()),
7 (1004, 'William', 'Sanchez', DATE '1970-08-01', 'Male', 'United Kingdom', 'Turkey', 'P47295268', NOW(), NOW()),
8 (1005, 'Robert', 'Miller', DATE '2008-03-13', 'Male', 'France', 'United Kingdom', 'P22981052', NOW(), NOW()),
9 (1006, 'Barbara', 'Ramirez', DATE '2000-11-09', 'Female', 'Kazakhstan', 'Japan', 'P81971316', NOW(), NOW()),
10 (1007, 'Patricia', 'Robinson', DATE '2003-12-17', 'Male', 'Canada', 'Poland', 'P94374605', NOW(), NOW()),
11 (1008, 'Karen', 'Lewis', DATE '2002-06-12', 'Male', 'France', 'Kazakhstan', 'P98753260', NOW(), NOW()),
12 (1009, 'Linda', 'Harris', DATE '1995-12-18', 'Male', 'Brazil', 'France', 'P61019678', NOW(), NOW()),
13 (1010, 'William', 'Anderson', DATE '2002-09-24', 'Male', 'Turkey', 'Turkey', 'P28119557', NOW(), NOW()),
14 (1011, 'Nancy', 'Rodriguez', DATE '1976-05-28', 'Male', 'Canada', 'Brazil', 'P31931511', NOW(), NOW()),
15 (1012, 'Joseph', 'Gonzalez', DATE '1994-03-21', 'Male', 'Turkey', 'Kazakhstan', 'P40742311', NOW(), NOW()),
16 (1013, 'Mark', 'Johnson', DATE '1998-04-20', 'Female', 'Poland', 'France', 'P38317637', NOW(), NOW()),
17 (1014, 'Sandra', 'Jackson', DATE '1998-03-26', 'Male', 'Japan', 'United Kingdom', 'P96282117', NOW(), NOW()),
18 (1015, 'Joseph', 'Jones', DATE '1993-10-06', 'Male', 'Brazil', 'Canada', 'P82340307', NOW(), NOW()),
19 (1016, 'William', 'White', DATE '2008-06-08', 'Female', 'Turkey', 'Brazil', 'P28566572', NOW(), NOW()),
20 (1017, 'Thomas', 'Thomas', DATE '1978-02-26', 'Male', 'France', 'Germany', 'P94214302', NOW(), NOW()),
21 (1018, 'Jennifer', 'Sanchez', DATE '2007-11-15', 'Male', 'United Kingdom', 'United Kingdom', 'P89978709', NOW(), NOW()),
22 (1019, 'Joseph', 'Taylor', DATE '1992-07-22', 'Male', 'France', 'Canada', 'P45812670', NOW(), NOW()),
23 (1020, 'Anthony', 'Lee', DATE '2000-07-08', 'Male', 'Poland', 'United Kingdom', 'P31227574', NOW(), NOW()),
24 (1021, 'Joseph', 'Smith', DATE '1993-08-18', 'Male', 'Canada', 'France', 'P93926371', NOW(), NOW()),
25 (1022, 'Elizabeth', 'Clark', DATE '2015-07-19', 'Male', 'Germany', 'Turkey', 'P31682744', NOW(), NOW()),
26 (1023, 'Sarah', 'Harris', DATE '2017-07-31', 'Male', 'Italy', 'Turkey', 'P75579548', NOW(), NOW()),
27 (1024, 'James', 'Brown', DATE '2002-07-26', 'Female', 'Brazil', 'Kazakhstan', 'P42329237', NOW(), NOW()),
28 (1025, 'Donald', 'Jackson', DATE '1977-01-24', 'Male', 'Japan', 'France', 'P81498611', NOW(), NOW()),
29 (1026, 'Anthony', 'Jones', DATE '1981-07-09', 'Female', 'Canada', 'Germany', 'P45575298', NOW(), NOW()),
30 (1027, 'Thomas', 'Ramirez', DATE '2007-12-19', 'Male', 'Canada', 'Brazil', 'P51837852', NOW(), NOW()),
31 (1028, 'Richard', 'Perez', DATE '2003-07-03', 'Female', 'Canada', 'Japan', 'P26246908', NOW(), NOW()),
32 (1029, 'Linda', 'Davis', DATE '1975-09-30', 'Female', 'Kazakhstan', 'Italy', 'P84346688', NOW(), NOW()),
33 (1030, 'Linda', 'Jackson', DATE '1989-10-04', 'Male', 'France', 'Kazakhstan', 'P40728048', NOW(), NOW()),
34 (1031, 'John', 'Lewis', DATE '1972-10-26', 'Female', 'France', 'Canada', 'P41044441', NOW(), NOW()),
35 (1032, 'William', 'Perez', DATE '2013-07-20', 'Male', 'Canada', 'Germany', 'P86044106', NOW(), NOW()),
36 (1033, 'Charles', 'Thomas', DATE '1991-10-20', 'Female', 'United Kingdom', 'Brazil', 'P22668194', NOW(), NOW()),
37 (1034, 'Patricia', 'Perez', DATE '2008-09-02', 'Female', 'United Kingdom', 'United Kingdom', 'P72682989', NOW(), NOW()),
38 (1035, 'Margaret', 'White', DATE '1974-11-11', 'Male', 'Kazakhstan', 'United Kingdom', 'P55540424', NOW(), NOW()),
39 (1036, 'Betty', 'Ramirez', DATE '1979-10-21', 'Male', 'Brazil', 'Brazil', 'P81979855', NOW(), NOW()),
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.031'.

2.



The screenshot shows the pgAdmin 4 interface. The 'Query History' window is open, displaying a small SQL query that inserts one row into the 'airline' table. The query is as follows:

```
1 INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
2 VALUES (101, 'KZ', 'KazAir', 'Kazakhstan', NOW(), NOW());
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.048'. A green notification box at the bottom right states 'Query returned successfully in 48 msec.'.

3.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'airline' table selected under the 'Tables' folder. The main pane shows the 'airline' table's columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The Query editor contains the following SQL statement:

```
1 UPDATE airline
2 SET airline_country = 'Turkey'
3 WHERE airline_name = 'KazAir';
```

The Data Output pane shows the result of the query:

```
UPDATE 1
Query returned successfully in 31 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.031' and 'CRLF Ln 3, Col 31'.

4.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'airline' table selected under the 'Tables' folder. The main pane shows the 'airline' table's columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The Query editor contains the following SQL statement:

```
1 INSERT INTO airline (airline_id, airline_code, airline_name, airline_country, created_at, updated_at)
2 VALUES
3 (182, 'AE', 'AirEasy', 'France', NOW(), NOW()),
4 (183, 'FH', 'FlyHigh', 'Brazil', NOW(), NOW()),
5 (184, 'FF', 'FlyFly', 'Poland', NOW(), NOW());
```

The Data Output pane shows the result of the query:

```
INSERT 0 3
Query returned successfully in 38 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.030' and 'CRLF Ln 5, Col 1'.

5.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'passengers' table selected. The main query editor contains the following SQL code:

```
1 DELETE FROM flights
2 WHERE EXTRACT(YEAR FROM sch_arrival_time) = 2024;
```

The Data Output pane at the bottom shows the result of the query:

```
DELETE 0
Query returned successfully in 29 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.029' and 'CRLF Ln 2, Col 50'.

6.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'booking' table selected. The main query editor contains the following SQL code:

```
1 UPDATE booking
2 SET ticket_price = ticket_price * 1.15;
```

The Data Output pane at the bottom shows the result of the query:

```
UPDATE 150
Query returned successfully in 42 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.042' and 'CRLF Ln 2, Col 40'.

7.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'booking' table selected under the 'passengers' schema. The table has columns: passenger_id (integer), first_name (character varying), last_name (character varying), date_of_birth (date), gender (character varying(5)), country_of_citizenship (character), country_of_residence (character), passport_number (character), created_at (timestamp with time zone), and updated_at (timestamp with time zone). The main query editor on the right contains the following SQL query:

```
1 DELETE FROM booking
2 WHERE ticket_price < 10000;
```

The Data Output pane at the bottom shows the result of the query:

```
DELETE 9
Query returned successfully in 30 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.030' and 'CRLF Ln 2, Col 28'.

8.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays the database structure, with the 'airline' table selected under the 'passengers' schema. The table has columns: passenger_id (integer), first_name (character varying), last_name (character varying), date_of_birth (date), gender (character varying(5)), country_of_citizenship (character), country_of_residence (character), passport_number (character), created_at (timestamp with time zone), and updated_at (timestamp with time zone). The main query editor on the right contains the following SQL query:

```
1 UPDATE airline
2 SET airline_code = 'UNK'
3 WHERE airline_code IS NULL;
```

The Data Output pane at the bottom shows the result of the query:

```
UPDATE 0
Query returned successfully in 40 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.040' and 'CRLF Ln 3, Col 28'. A green checkmark icon and the text 'Query returned successfully' are visible in the bottom right corner.

9.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree view of the database schema. The 'passengers' table is expanded, showing its columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The 'baggage_check' table is also visible in the tree. The main query editor on the right contains the following SQL query:

```
1 DELETE FROM baggage_check
2 WHERE created_at < '2023-06-01'
3 AND check_result = 'Not checked';
```

The Data Output pane at the bottom shows the result of the query: 'DELETE 0' and 'Query returned successfully in 29 msec.' The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.029' and 'CRLF Ln 3, Col 36'.

10.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree view of the database schema. The 'passengers' table is expanded, showing its columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The 'airport' table is also visible in the tree. The main query editor on the right contains the following SQL query:

```
1 DELETE FROM airport
2 WHERE state IS NULL
3 AND city IN ('Mlawe', 'Kepuh');
```

The Data Output pane at the bottom shows the result of the query: 'DELETE 0' and 'Query returned successfully in 30 msec.' A green status bar at the bottom right indicates '✓ Query returned successfully in 30 msec. ✕'. The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.030' and 'CRLF Ln 3, Col 34'.

11.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with the 'passengers' table selected. The main pane shows the SQL query editor with the following query:

```

1 INSERT INTO baggage_check (baggage_check_id, check_result, created_at, updated_at, booking_id, passenger_id)
2 VALUES (5001, 'Not checked', NOW(), NOW(), 1, 1)
3 RETURNING baggage_check_id, created_at;

```

The 'Data Output' tab is active, showing the result of the query:

baggage_check_id	created_at
5001	2026-09-27 11:16:23.027282

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.042'.

12.

The screenshot shows the pgAdmin 4 interface. The left sidebar displays the database structure, with the 'passengers' table selected. The main pane shows the SQL query editor with the following query:

```

1 UPDATE airline
2 SET airline_country = UPPER(airline_country);

```

The 'Data Output' tab is active, showing the result of the query:

UPDATE 19
Query returned successfully in 42 msec.

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.042'. A green notification box at the bottom right states 'Query returned successfully in 42 msec.'.

13.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree view of the database schema. The 'airline' table is selected under the 'Tables' folder. The main pane shows the SQL editor with the following query:

```
1 UPDATE airline
2 SET airline_name = 'Global Airways',
3     airline_country = 'United Kingdom',
4     updated_at = CURRENT_DATE
5 WHERE airline_id = 5;
```

The Data Output pane at the bottom shows the result of the query:

```
UPDATE 1
Query returned successfully in 29 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 29 msec." The status bar also shows "CRLF" and "Ln 5, Col 22".

14.

The screenshot shows the pgAdmin 4 interface. On the left, the Object Explorer displays a tree view of the database schema. The 'airport' table is selected under the 'Tables' folder. The main pane shows the SQL editor with the following query:

```
1 UPDATE airport
2 SET state = 'Capital District'
3 WHERE city IN ('Astana', 'London', 'Tokyo');
```

The Data Output pane at the bottom shows the result of the query:

```
UPDATE 3
Query returned successfully in 28 msec.
```

A green status bar at the bottom right indicates: "Query returned successfully in 28 msec." The status bar also shows "CRLF" and "Ln 3, Col 45".

15.

The screenshot displays the pgAdmin 4 web interface. On the left, the 'Object Explorer' pane shows a hierarchical view of the database structure. The 'Tables (10)' folder is expanded, and the 'passengers' table is selected, showing its columns: passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship, country_of_residence, passport_number, created_at, and updated_at. The main pane shows a SQL query editor with the following SQL statement:

```
1 UPDATE baggage_check;
2 SET check_result = 'Checked';
3 WHERE check_result = 'Not checked';
4 AND created_at >= '2024-03-01';
5 AND created_at < '2024-04-01';
```

Below the query editor, the 'Data Output' tab is active, displaying the result of the query:

```
UPDATE 6
Query returned successfully in 29 msec.
```

A green notification box at the bottom right of the interface states: 'Query returned successfully in 29 msec. X'. The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.029' and 'CRLF Lin 5, Col 33'.